

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (CL)

December 2011

CL211 Environmental Engineering I

Date: 09.12.2011, Friday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Explain beneficial uses of Water Examination. [03]
(b) Define 'Environmental Engineering'. Write scope of Environmental Engineering. [04]
- Q - 2 (a) Draw the layout of surface water treatment plant. Write function of each unit and explain different types of settling. [07]
(b) Discuss chemical characteristics of water. [04]
(c) Explain process of coagulation. [03]

OR

- Q - 2 (a) Compare the characteristics of different coagulants used in water treatment. Explain the theory of filtration. [07]
(b) Classify Water Pollutants. What is 'Super Chlorination'? [04]
(c) Write importance of Indian Standards for Drinking Water. [03]
- Q - 3 (a) With neat sketch explain the working of Rapid Sand Filter. [07]
(b) Design a circular sedimentation tank for water works which has to supply daily 3 million liters of water. The detention period in the tank for water for 4 hours and depth of water in the tank is 3.5m. [04]
(c) What is hardness? Give classification of hardness. [03]

OR

- Q - 3 (a) Explain the working of pressure Filter. Explain methods of disinfection in detail. [07]
(b) Define Eutrofication. Enlist design considerations of water treatment plant. [04]
(c) Differentiate between Slow Sand filter and Rapid Sand Filter. [03]

SECTION - II

- Q - 4 (a) Enlist objectives of Water Supply System. Define "per capita demand." [03]
(b) Which are the factors affecting per capita demand? Explain hourly variation in water demand. [04]
- Q - 5 (a) What is the purpose of intake structure? Explain any one type of intake structure with neat sketch. [07]
(b) Give classification of water distribution systems. Explain arithmetic increase method for population forecasting. [04]
(c) Define the term 'Ventilation'. What are the objectives of ventilation system? [03]

OR

- Q - 5 (a) Draw the neat sketch showing the house service connection from the distribution main and state the function of each component. [07]
(b) Define the term 'Cavitation'. Enlist effects of cavitation. [04]
(c) Explain natural system of ventilation. [03]
- Q - 6 (a) Enlist types of traps used in house drainage. Explain gully trap in detail. [07]
(b) Explain methods of refuse disposal. [07]

OR

- Q - 6 (a) Give importance of air conditioning and lighting in a ventilation system. [07]
(b) Explain different sanitary fittings in house drainage system. [07]
